

ANIMESH SAH

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RESEARCH INTERESTS

Model-independent tests of the standard cosmological model; the impact of cosmic anisotropies, inhomogeneities, and bulk flows on cosmological inference; peculiar velocities; bulk flows; and the application of machine learning methods to large-scale structure analysis.

EDUCATION

Ph.D., Physics 2023 – Present (*expected early 2027*)

Tata Institute of Fundamental Research (TIFR), Mumbai, India. Advisor: Dr. Mohamed Rameez.

M.Sc., Physics Sept. 2020 – 2023

Tata Institute of Fundamental Research (TIFR), Mumbai, India. (Integrated Ph.D. programme.)

B.Sc. (Honours), Physics July 2017 – June 2020

University of Delhi, India.

JOURNAL PUBLICATIONS

1. **A. Sah** et al., “Anisotropy in Pantheon+ Supernovae,” published in [European Physical Journal C](#) [[arXiv:2411.10838](#)].
2. **A. Sah** et al., “Pantheon+ supernovae corrected for progenitor age indicate the universe is decelerating,” published in [Monthly Notices of the Royal Astronomical Society](#) [[arXiv:2606.09650](#)].

CONFERENCE PROCEEDINGS

1. **A. Sah** et al., “Examining the Cosmic Anisotropy with the Pantheon+ Supernovae,” published in [Springer Nature Proceedings](#).

RESEARCH EXPERIENCE

Anisotropy in the Pantheon+ Supernovae Compilation 2023–2025

with Dr. Mohamed Rameez, TIFR; Prof. Subir Sarkar, University of Oxford; and Prof. Christos Tsagas, Aristotle University of Thessaloniki

- Found statistically significant evidence for anisotropy in the Pantheon+ Type Ia supernova dataset using Maximum Likelihood Estimation.

Tests of Tilted Cosmology 2025–Present

with Dr. Mohamed Rameez, TIFR

- Leading the development of model-independent pipeline to test tilted Λ CDM cosmologies against upcoming LSST supernova data with realistic simulations from SNANA.

Mock catalogs for the DESI Fundamental Plane sample 2025 – Present

with Dr. Shadab Alam, TIFR

- Building realistic mock catalogs for a photometric sample of $\sim 480K$ early-type galaxies ($z < 0.15$) from the DESI Legacy Imaging Survey, using the AbacusSummit N-body simulation suite as the underlying dark-matter field.

TALKS

The Concurrence of Mega-Surveys International Centre for Theoretical Sciences (ICTS), Bangalore (August 2026)

Presented a flash talk

Supernova Cosmology: New Challenges and Opportunities

Yonsei University, Seoul (June 2026)

Talk on Pantheon+ anisotropy and progenitor-age systematics.

Cosmic Lighthouses: Astrophysical and Cosmological Challenges with Type Ia Supernovae

Kavli

Institute for Cosmology, Cambridge (July 2025)

Talk on Pantheon+ anisotropy

5th EuCAPT Annual Symposium

Online (May, 2025)

[Talk](#) on Anisotropy in Pantheon+ supernovae

XXVI DAE-BRNS High Energy Physics Symposium 2024

Banaras Hindu University, Varanasi (December 2024)

Contributed talk on cosmic anisotropy with Pantheon+.

A Multipolar Universe?

Aristotle University of Thessaloniki, Greece (September 2023)

Talk on Pantheon+ anisotropy.

Less Travelled Path to the Dark Universe

International Centre for Theoretical Sciences (ICTS), Bangalore (March 2023)

[Talk](#) on cosmological analysis with the Pantheon+ compilation.

SCHOOLS & WORKSHOPS

The Concurrence of Mega-Surveys

International Centre for Theoretical Sciences (ICTS), Bangalore

(July-August 2026)

Project: implemented and trained a transformer-based decoder (GOTHAM) to accelerate halo catalogue generation from particle-mesh simulations via distributed multi-GPU training.

Rethinking Cosmology

Raman Research Institute, Bengaluru (December 2025)

Largest Cosmological Surveys and Big Data Science

International Centre for Theoretical Sciences (ICTS),

Bangalore (May 2023)

Project: reanalyzed Dark Energy Survey cosmic-shear measurements using machine-learning techniques.

TECHNICAL SKILLS

- **Statistical methods:** Frequentist and Bayesian inference, Maximum Likelihood Estimation, MCMC, Hamiltonian Monte Carlo (HMC) (emcee, NumPyro).
- **High-performance computing:** Job submission, scheduling, and parallelization across HTCondor, SLURM, and PBS clusters; GPU-accelerated computing.
- **Programming:** Python (NumPy, SciPy, Astropy, Pandas, JAX), C++, Mathematica.
- **Machine learning:** TensorFlow, PyTorch, gradient-boosted trees, Transformers, CNNs, distributed multi-GPU training.
- **Cosmology software & data products:** SNANA, SALT2/SALT3, TreeCorr, HEALPix.
- **Other:** Git/GitHub, LaTeX, large-scale FITS/HDF5 data handling.

PROFESSIONAL MEMBERSHIPS

- Member, LSST Dark Energy Science Collaboration (DESC).

REFERENCES

Available upon request.